

PBA Final Report - Data processing & analysis

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1 Data processing

1. Read the data

```
exp_data <- read_csv(url("https://drive.google.com/uc?export=download&id=1estf46syRbfG2K8JfW9myfTq2qDp0"))
```

```
## Rows: 77 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (3): comment_behavior, gender, age_group
## dbl (7): id, treatment, control, hotel1, hotel2, hotel3, comment_count
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

2. Select and rename the columns

```
df <- exp_data |>
  select('id', 'treatment', 'hotel1', 'hotel2', 'hotel3') |>
  pivot_longer(
    cols = c('hotel1', 'hotel2', 'hotel3'),
    names_to = 'hotel',
    values_to = 'score'
  ) |>
  mutate(
    scenario = case_when(
      hotel == "hotel1" ~ "negative",
      hotel == "hotel2" ~ "neutral",
      hotel == "hotel3" ~ "positive"
    )
  )
```

```
)
)
df
```

```
## # A tibble: 231 x 5
##       id treatment hotel  score scenario
##   <dbl>      <dbl> <chr>  <dbl> <chr>
## 1     1         0 hotel1     7 negative
## 2     1         0 hotel2     6 neutral
## 3     1         0 hotel3     8 positive
## 4     2         0 hotel1     5 negative
## 5     2         0 hotel2     7 neutral
## 6     2         0 hotel3     6 positive
## 7     3         1 hotel1     4 negative
## 8     3         1 hotel2     5 neutral
## 9     3         1 hotel3     8 positive
## 10    4         0 hotel1     4 negative
## # i 221 more rows
```

3. Check the data completeness

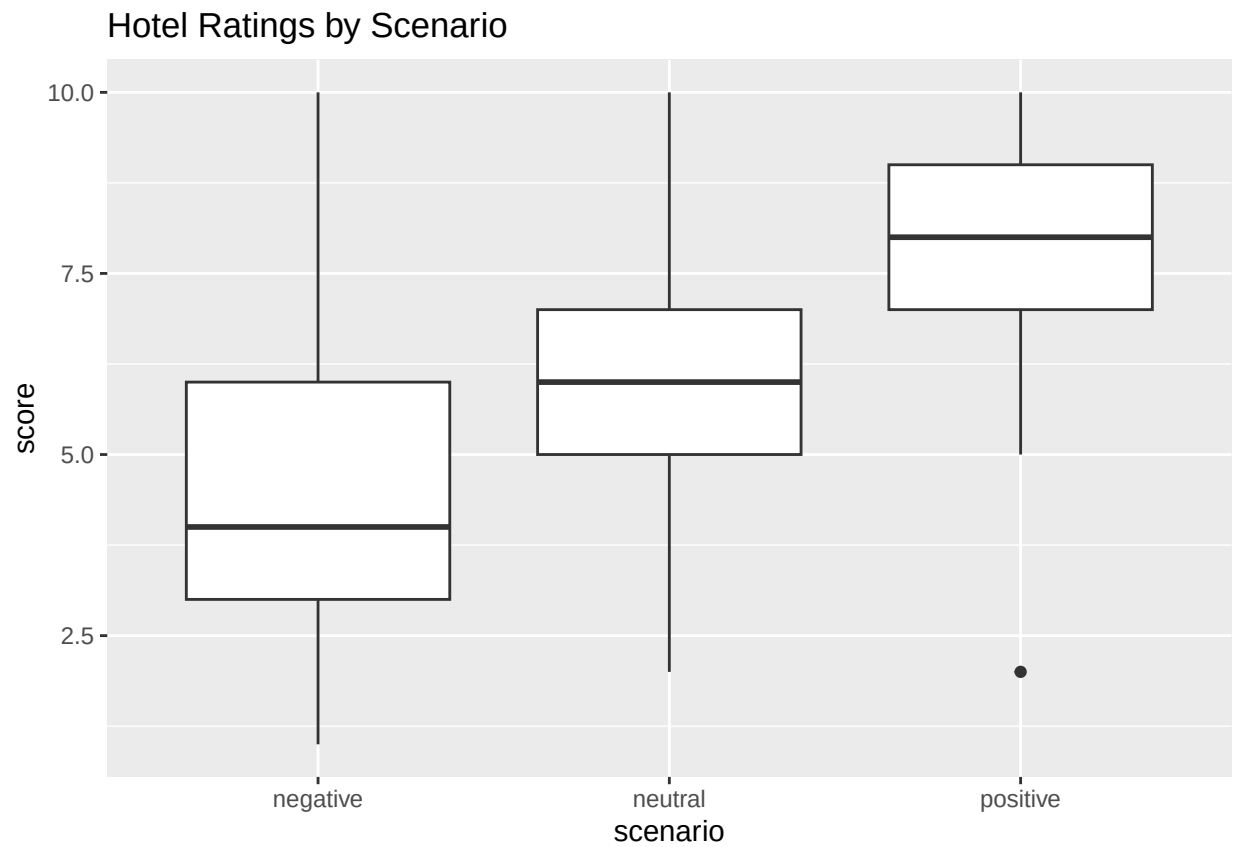
```
table(df$id)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
##  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3
## 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
##  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3
## 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77
##  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3  3
```

2 EDA

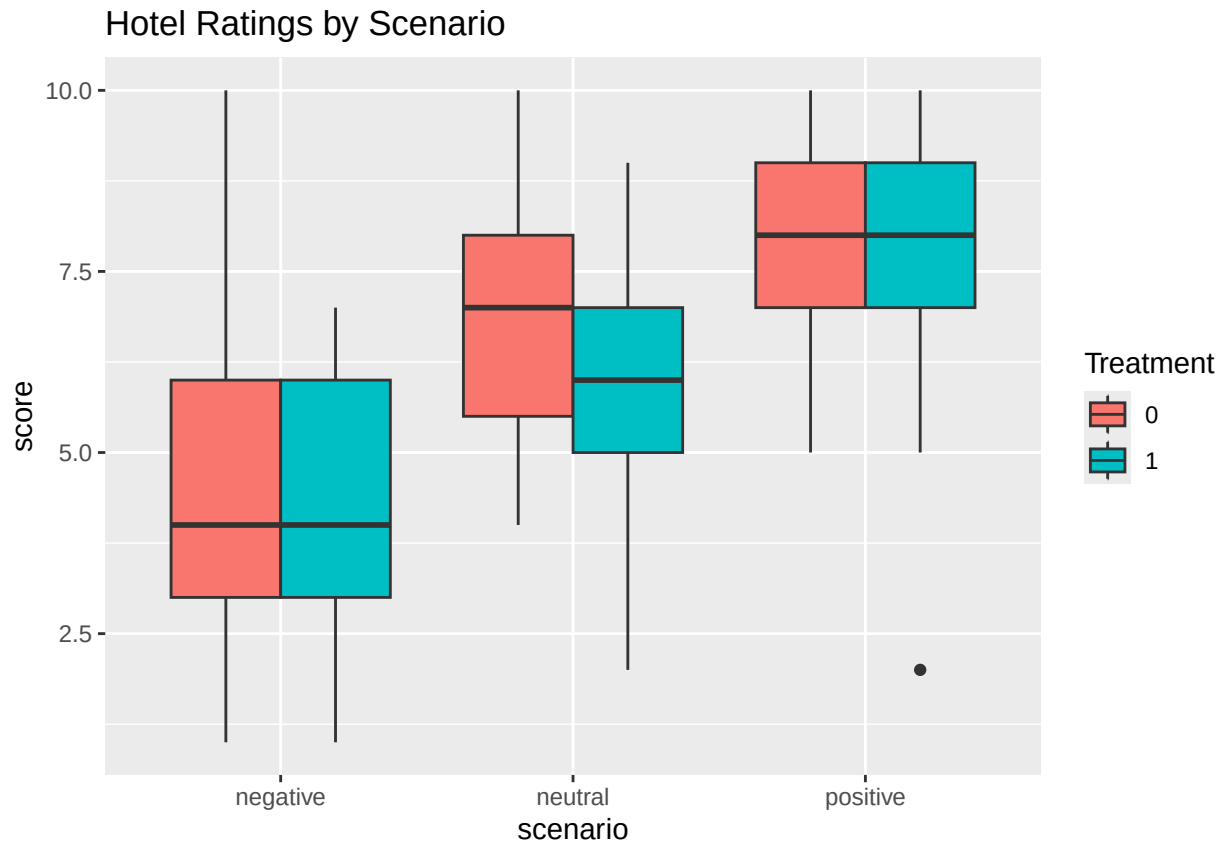
1. Boxplot

```
ggplot(df, aes(x = scenario, y = score)) +
  geom_boxplot() +
  labs(title = "Hotel Ratings by Scenario")
```



2. Boxplot with treatment

```
ggplot(df, aes(x = scenario, y = score, fill = factor(treatment))) +  
  geom_boxplot(position = position_dodge()) +  
  labs(title = "Hotel Ratings by Scenario", fill = "Treatment")
```



3 Linear Mixed Model Analysis

1. Set the 'neutral' scene as the baseline.

```
df$treatment <- factor(df$treatment)
df$scenario <- factor(df$scenario,
  levels = c("neutral", "positive", "negative"))
```

2. Model

```
m1 <- lmer(
  score ~ treatment * scenario + (1 | id),
  data = df
)
```

3. Statistic summary

```
summary(m1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ treatment * scenario + (1 | id)
```

```
## Data: df
##
## REML criterion at convergence: 859
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.48248 -0.53482  0.00796  0.60812  2.58815
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## id       (Intercept) 1.048    1.024
## Residual                1.706    1.306
## Number of obs: 231, groups: id, 77
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      6.6154    0.2657 174.4666  24.894 < 2e-16 ***
## treatment1      -0.5628    0.3783 174.4666  -1.488   0.139
## scenariopositive  1.2051    0.2958 150.0000   4.074 7.45e-05 ***
## scenarionegative -2.4615    0.2958 150.0000  -8.322 4.93e-14 ***
## treatment1:scenariopositive  0.5054    0.4210 150.0000   1.200   0.232
## treatment1:scenarionegative  0.6457    0.4210 150.0000   1.534   0.127
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) trtmn1 scnrps scnrng trtmnt1:scnrp
## treatment1    -0.703
## scenaripstv   -0.557  0.391
## scenaringtv   -0.557  0.391  0.500
## trtmnt1:scnrp  0.391 -0.557 -0.703 -0.351
## trtmnt1:scnrn  0.391 -0.557 -0.351 -0.703  0.500
```

4. Fixed effect

```
tidy_m1 <- tidy(m1, effects = "fixed")
kable(tidy_m1, digits = 3)
```

effect	term	estimate	std.error	statistic	df	p.value
fixed	(Intercept)	6.615	0.266	24.894	174.467	0.000
fixed	treatment1	-0.563	0.378	-1.488	174.467	0.139
fixed	scenariopositive	1.205	0.296	4.074	150.000	0.000
fixed	scenarionegative	-2.462	0.296	-8.322	150.000	0.000
fixed	treatment1:scenariopositive	0.505	0.421	1.200	150.000	0.232
fixed	treatment1:scenarionegative	0.646	0.421	1.534	150.000	0.127

5. Random effect

```
tidy_m1_random <- tidy(m1, effects = "ran_pars")
kable(tidy_m1_random, digits = 3)
```

	x		x
Conditional R2	0.657	Marginal R2	0.446

effect	group	term	estimate
ran_pars	id	sd__(Intercept)	1.024
ran_pars	Residual	sd__Observation	1.306

6. Check the Model fit

```
r2 <- r2(m1)
kable(r2, digits = 3)
```

4 Bootstrap resampling for the neutral hotel scenario

1. Bootstrap

```
df_neutral <- df %>% filter(scenario == "neutral")
ate_boots <- df_neutral |>
  rep_slice_sample(prop = 1, reps = 1000, replace = TRUE) |>
  group_by(replicate, treatment) |>
  summarize(mean_score = mean(score, na.rm = TRUE), .groups = "drop") |>
  pivot_wider(
    names_from = treatment,
    values_from = mean_score
  ) |>
  mutate(ate = `1` - `0`);ate_boots
```

```
## # A tibble: 1,000 x 4
##   replicate '0' '1'   ate
##       <int> <dbl> <dbl> <dbl>
## 1         1  6.55  5.86 -0.685
## 2         2  6.23  6.12 -0.110
## 3         3  6.74  6.51 -0.224
## 4         4  6.44  6.28 -0.161
## 5         5  6.54  6.10 -0.438
## 6         6  6.94  6.27 -0.676
## 7         7  6.47  6.03 -0.436
## 8         8  6.92  6.1  -0.819
## 9         9  6.89  5.85 -1.04
## 10        10  6.83  6    -0.829
## # i 990 more rows
```

2. Calculate the 95% confidence interval

```
boot_ci <- quantile(ate_boots$ate, c(0.025, 0.975))

boot_ci
```

```
##          2.5%          97.5%
## -1.16546964  0.07497076
```

3. Visualize the ate distribution

```
ggplot(ate_boots, aes(x=ate)) +  
  geom_histogram(aes(y = after_stat(density)), color = "white") +  
  geom_vline(xintercept = unlist(boot_ci), linewidth = 1)
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

